Sample:

We show the workflow from making input data to visualization using the data of Cloncurry MT.
Please try to use your own data if you have.

1.Making Input Data from EDI file.

a) download EDI file from https：//geoscience.data.qld.gov.au/data/dataset/mt099997/resource/geo-doc1255018-mt099997
b) make "edilist.lst" file in which EDI file names are written (see figure. 1)
c) Execute readEDI,py, which should be located in the same folder as EDI files. Please set EPGS coord systems, frequencies, and errorfloor. The frequencies are interpolated in log10 scale (Figure. 2).
NOTE: This Program Uses $+i\omega t$ Systems!!!!!!!
d) If you don't have enough modules in Python, please install with "pip".
e) The used frequencies are written in "CalcFreq.txt". We use this later.
f) obsDataImpedance.txt and obsDataTipper.txt are input files.

```
L02S18.edi
L02S19.edi
L03S02.edi
L03S03.edi
L03S04.edi
L03S18.edi
L03S19.edi
L03S20.edi
L03S21.edi
L03S21R.edi
L04S021.edi
L04S03.edi
L04S04.edi
L04S05R.edi
L04S06.edi
L04S07.edi
L04S08.edi
L04S09R.edi
L04S12.edi
L04S13.edi
L04S14.edi
L04S15.edi
L04S16.edi
L04S17R.edi
L04S18.edi
L04S19.edi
L04S20-1.edi
L04S20-2.edi
```

Figure. 1

```python
import re
import copy
from pyproj import Transformer
import matplotlib.pyplot as plt
from scipy import interpolate
#===Parameters====================
ediFileList="edilist.lst"
outputImpedanceFile="obsDataImpedance.txt"
outputTipperFile="obsDataTipperFile.txt"
outputFreqFile="CalcFreq.txt"
outputLocationFile="ObsStationLocation.txt"
wgs84_epsg, rect_epsg = 4326,  3112  #3112  is for australia

minFreq=0.001 #For Calculation
maxFreq=100
numOfFreq=16
efZxx=0.1
efZxy=0.05
efZyx=0.05
efZyy=0.1
efTx=0.1
efTy=0.1
needs_SIconv=1

#==================================
```

Figure. 2

The format of Impedance file is as below:

[x coord] [y coord] [obs file name]

[ZxxR] [ZxxI] [ZxyR] [ZxyI] [ZyxR] [ZyxI] [ZyyR] [ZyyI]

[RZxxR] [RZxxI] [RZxyR] [RZxyI] [RZyxR] [RZyxI] [RZyyR] [RZyyI]

Here, RZ means standard deviation (Figure. 3).


```
11633.048471272574 27919.594101383816 .¥.¥L02S18.edi
-7.901235727374723e-05 7.315036880595087e-05 -0.00024858435478058896 -6.794061712008636e-05 -0.00023873664654343877 -0.00013904159057690177 0.0009250071758232608 -7.984142007852135e-05
0.0004977402052854487 0.0004977402052854487 0.0006832359430001499 0.0006832359430001499 0.0005844487071457267 0.0005844487071457267 0.0008049515941108572 0.0008049515941108572
-0.00020149503913187517 0.00017668031716502994 5.662020359370318e-05 2.4404414823272444e-05 -0.0009162480354911032 -0.0005629576506662632 0.0010171558901576926 0.0004024308106622822
0.0004557652632826498 0.0004557652632826498 0.00026423954486164033 0.00026423954486164033 0.0005780567430639442 0.0005780567430639442 0.0003270668463373448 0.0003270668463373448
-5.5619507712493586e-05 -0.00012590925701835045 0.0002456444895020837 0.00032509300436506374 -0.0005749704405136349 -0.0008208039575696095 0.0009048447723991273 0.00032202016498398975
0.0003906151424080009 0.0003906151424080009 0.00034146195782247123 0.00034146195782247123 0.0003158798053195448 0.0003158798053195448 0.00027635099389594875 0.00027635099389594875
0.00029595326737653964 0.00022414890755349164 0.0003764145663236413 0.0003935776904605979 -0.0011658702261095285 -0.0015876432212291965 0.0010607383526757045 0.0005210365718090048
0.0004068232508261386 0.0004068232508261386 0.0004201003048957029 0.0004201003048957029 0.00033566622364327527 0.00033566622364327527 0.0003433434385051819 0.0003433434385051819
0.0007800898296143768 0.0005369239364645493 -6.963038074843904e-05 0.0005578559996196727 -0.001887061684384837 -0.002169562094112001 0.001446944390881251 0.0005205495292445112
0.0004170014643093185 0.0004170014643093185 0.000541995194215011 0.000541995194215011 0.0003227556547969941 0.0003227556547969941 0.00042011573422963454 0.00042011573422963454
0.0009459526415026562 0.0006751694761230422 0.0007974364628119116 0.001381475631134972 -0.0029147217089633905 -0.0031641523559093974 0.0018680684831973613 -0.0007729378984906939
0.0003835080154995095 0.0003835080154995095 0.0008044509215420394 0.0008044509215420394 0.0009426216511824553 0.0009426216511824553 0.0020066119965811468 0.0020066119965811468
0.0013317404445341522 0.000978521603825153 0.0015591925524799902 0.0018502308733966129 -0.003330011107686188 -0.0039451487130948626 -0.0014139352559543875 -0.0007062823153969804
0.0021094592795926864 0.0021094592795926864 0.0023563422265850194 0.0023563422265850194 0.006269330529516071 0.006269330529516071 0.007015396502911375 0.007015396502911375
0.004560457818029409 0.0008166557808197712 0.0011116383645858553 0.0028982556495260243 -0.01856660633154239 -0.005917956153374001 0.0040706768900930935 -0.0038383054715529735
0.005412682496605047 0.005412682496605047 0.0022253229779335107 0.0022253229779335107 0.021591438593577438 0.021591438593577438 0.008876880309760221 0.008876880309760221
0.0027885549165455607 0.0014193475029277796 0.002957399079909252 0.0035535212202381775 -0.010421961184192476 -0.007176863734669201 0.0010500942580450186 -0.0022631153637840276
0.0007648646474986962 0.0007648646474986962 0.0003824323237493481 0.0003824323237493481 0.0003824323237493481 0.0003824323237493481 0.0007648646474986962 0.0007648646474986962
0.0037120795996690233 0.0011460875585913098 0.005064783991924953 0.00557869641403819 -0.01474049607108337 -0.008041184233874916 -6.918158198551276e-05 -0.0031941751351653594
0.0011248055290711749 0.0011248055290711749 0.0005624027645355874 0.0005624027645355874 0.0005624027645355874 0.0005624027645355874 0.0011248055290711749 0.0011248055290711749
0.004436155712355075 0.0006261361580705426 0.008133339420288762 0.0073876075694158634 -0.0187840283260884 -0.007410353985638785 -0.0024976120581212537 -0.0032178717088817642
0.0014895369145733262 0.0014895369145733262 0.0007447684572866631 0.0007447684572866631 0.0007447684572866631 0.0007447684572866631 0.0014895369145733262 0.0014895369145733262
0.0047114821156587165 -0.00045777264396568737 0.01267628249020406 0.009195304804356196 -0.022604275885069816 -0.006472075438404297 -0.004334171361898455 -0.0018868129830786797
0.0019188838170518263 0.0019188838170518263 0.0009594419085259131 0.0009594419085259131 0.0009594419085259131 0.0009594419085259131 0.0019188838170518263 0.0019188838170518263
0.004188104625773801 -0.001837630796677017 0.01690881693406226 0.010704790019882704 -0.024257768469420456 -0.006368884408542379 -0.004725900153466137 -0.0005138878673190527
0.002240339355202095 0.002240339355202095 0.0011201696776010476 0.0011201696776010476 0.0011201696776010476 0.0011201696776010476 0.002240339355202095 0.002240339355202095
0.002728319582388041 -0.0030996695791471293 0.02131150865724327 0.013312145099316367 -0.025590465151319353 -0.008512386628538869 -0.004508967051859759 0.0006263021540554948
0.002603204663500203 0.002603204663500203 0.0013016023317501016 0.0013016023317501016 0.0013016023317501016 0.0013016023317501016 0.002603204663500203 0.002603204663500203
0.0005593129097990615 -0.0037661605794247974 0.026174435390611855 0.018290802093143682 -0.02749180018135323 -0.01425243206269482 -0.0033710770384785672 0.0013311952687446913
0.00314456147189761 0.00314456147189761 0.001572280735948805 0.001572280735948805 0.001572280735948805 0.001572280735948805 0.00314456147189761 0.00314456147189761
```


Figure. 3

The format of Tipper file is as below:

[x coord] [y coord] [obs file name]

[TxR] [TxII] [TyR] [TyI]

[RTxR] [RTxI] [RTyR] [RTyI]

Here, RT means standard deviation (Figure. 4).

```
11633.048471272574 27919.594101383816 .¥.¥L02S18.edi
-0.024527525369015223 0.0020609783371728536 -0.3767424666152214 0.2361394357092865
0.33009060030581455 0.33009060030581455 0.4479274781743163 0.4479274781743163
0.03172174609976668 0.07270692022897246 -0.24784646877000105 0.07607694326030516
0.24714980043860724 0.24714980043860724 0.14649112126256364 0.14649112126256364
-0.06457218263352277 0.06557813444705979 -0.10957408720799207 0.1447633458603041
0.17617554492526094 0.17617554492526094 0.15395537923801997 0.15395537923801997
0.11701563128968076 0.08111461212834388 -0.12326543346335492 0.05432322704347829
0.12462692467236115 0.12462692467236115 0.12676791787495018 0.12676791787495018
0.12853200009213264 -0.027370082750757026 -0.05891122082987102 0.1720295877280024
0.12270113684091859 0.12270113684091859 0.15971409163374412 0.15971409163374412
0.15056466625466455 -0.17188382279757508 -0.3173929806148822 0.3966138494252348
0.592326447753032 0.592326447753032 1.2910515599504238 1.2910515599504238
0.42244217151260305 0.2624296252567535 -0.3999998925701589 -0.1798177648359841
2.6107087529312776 2.6107087529312776 2.9688976215508864 2.9688976215508864
1.6888479860151775 1.9455235963975832 0.15335658823470086 -0.9884540846849525
7.6093301447633515 7.6093301447633515 3.130513148664485 3.130513148664485
-0.11792953385381094 -0.6126843831754535 0.08208521105559943 -0.016322981072605047
1.3078652817226055 1.3078652817226055 1.1663347242905135 1.1663347242905135
-0.12049628143525631 -0.014820202859147343 -0.23856740911490376 -0.04311068743962425
0.27194340058629435 0.27194340058629435 0.2865456448254374 0.2865456448254374
-0.2336736739090974 -0.11166893760953958 0.06442284312423577 0.38247219730591897
0.8933183241732614 0.8933183241732614 0.6241194362471738 0.6241194362471738
-0.1460659785563364 0.08443968081074568 0.18334942897306056 0.2931602453577469
0.36426250514240965 0.36426250514240965 0.3040329655249627 0.3040329655249627
-0.027272473130384625 0.03762441070130056 -0.07436820885606613 0.09272756031091706
0.04298546363099822 0.04298546363099822 0.07976302053620794 0.07976302053620794
0.008845155691670315 -0.014756185571726735 0.017261510139489206 0.03571710624460551
0.002612430602335599 0.002612430602335599 0.00401564011868978 0.00401564011868978
-0.004865078633946232 -0.00821634307471121 0.018502003242660766 0.018107792344843998
0.0020349379485142772 0.0020349379485142772 0.002707712788910197 0.002707712788910197
-0.0033786606887549293 -0.0030772986607567753 0.019660793131891757 0.016521107817 86193
0.002637303436594 68 0.00263730343659468 0.00267406352637551 2 0.00267406352637551 2
13576.485422696336 27868.70893485658 .¥.¥L02S19.edi
-0.16592594493949997 -0.027041937132206793 -0.2135743828002189 -0.14464346694396954
```

Figure. 4

2. Making Topography Data

You need to make Topography data as below:

[latitude] [longitude] [elevation]

Here, if the topography is below sea, you should set the minus values.

When the values are minus, the mesher recognize them as under sea.

We make mesh data using the interoilation of the topography data (Figure. 5).

```
-40.0 120.0 -4822.0
-39.99583333333333 120.0 -4835.0
-39.99166666666667 120.0 -4850.0
-39.9875 120.0 -4863.0
-39.983333333333334 120.0 -4876.0
-39.979166666666664 120.0 -4889.0
-39.975 120.0 -4902.0
-39.97083333333333 120.0 -4913.0
-39.96666666666667 120.0 -4926.0
-39.9625 120.0 -4937.0
-39.958333333333336 120.0 -4948.0
-39.954166666666666 120.0 -4959.0
-39.95 120.0 -4969.0
-39.94583333333333 120.0 -4978.0
-39.94166666666667 120.0 -4986.0
-39.9375 120.0 -4993.0
-39.93333333333333 120.0 -4999.0
-39.92916666666667 120.0 -5003.0
-39.925 120.0 -5007.0
-39.920833333333334 120.0 -5010.0
-39.916666666666664 120.0 -5010.0
-39.9125 120.0 -5010.0
```

3. Making Input Data

We use makeBoxData_MT_XYZ_INV_ForAustoralia.py to make an input file.
You need to set parameters in the .py file, and run the script as below:
python makeBoxData_MT_XYZ_INV_ForAustoralia.py
You will have input file "calcData.txt"
You need to set parameters below which is written in calcData.txt:

- #====PARAMETERS FOR MESSHER=====================
- #===========PARAMETERS FOR FREQENCIES====================
- #=============PARAMETERS FOR INVERSION===================

The meanings of parameter are as below:

i. PARAMETERS FOR MESSHER

numSplitPrecise ：Our program can divide mesh where we need precise mesh like "Adaptive mesh refinement (AMR)". You set the maximum layer of the AMR.
If you do not need AMR, please set this 0.

nx, ny, nz ：The number of divisions for each direction, X,Y, and Z.
nAir: The number of layers for Air.
thicknessAirLayer: The minumum thickness of Air [m].
minDx,minDy,minDz: The minumum length of cells for each directions [m].
dxMax, dyMax, dzMax:Maximum length of the cells for each direction [m].
dzMaxInAir:Maximum length of the air cells for each direction.
corenx, coreny, corenz: The numbers of cells where we keep minimum length, minDx,minDy,minDz.
corenz_air：The numbers of cells where we keep minimum length for ground to above (that is, air layer).
x_coeff, y_coeff, z_coeff, z_coeffAir: The ratio to get larger out of core zones described above.
topoData ：The file name made in 2. Making Topography Data.

latc, longc: The model center latitude and longitude. When you use "readEdi.py" in 1.Making Input Data from EDI file.", the the values are written in CenterLatLong.txt.

rect_epsg: You set coordinate systems from EPSG. You must set the same value as 1.Making Input Data from EDI file.

initResis: The initial resistivity value for subsurface (Ohm m).

oceanResis: Ocean Resistivity (Ohm m), Note that The cells assigned this value is fixed through the inversion.

airResis: Air Resistivity (Ohm m), Note that The cells assigned this value is fixed through the inversion.

xPrecise, yPrecise, zPrecise: You set the range of cell number from the center to each direction where the AMR should be assigned. It is set as comment out because we do not use them in this case.

ii. PARAMETERS FOR FREQENCIES

You set the frequencies used in the incersion.
You must set them corresponding with 1.Making Input Data from EDI file.
You can check the values in "CalcFreq.txt".

iii. PARAMETERS FOR INVERSION

You set parameters for the inversion.

UseGD: If you use this value as true. "Adamax" is used. If "False", L-BFGS is used. However, the developer experience recommend "True" for real data for robust convergence.

toleranceIterativeSolver: The tolerance for iterative solver.

maxIterationBiCGSTAB: The maximum number of iteration for iterative solver (actually, the solver is BiCGSafe, not BiCGSTAB though...).

UseIterativeSolver: Use iterative solver or not. !!!!YOU MUST SET THIS VALUE "TRUE"!!!! The program cannot accept "False" for now.

UseDistanceInModelConstraint: The flag for the model constaint term in the objective function. If "True", the constraint becomes weeker as the cells large, that is, the distance of the center of cells becomes large. If "False", the program contrant equally for each adjacent cell.

modifyGradient: The program uses the gradient vector for model update. In the case, the cells near surface tend to fluctuate largely. This flag is used for mitigating this phenomena. Please see the detail in the paper below: "3D magnetotelluric inversion using a limited-memory quasi – Newton optimization" Avdeev and Avdeeva (2009).

safetyFactor: The program adopts "Divergence free" method for quick convergence in iterative solver following Divergence correction を Dong and Egbert (2019). However, experimentally, the error of foward calclation becomes large when we directly adopt their method. Hence, we adopted this value for the precision. The calculation is faster but more rough especially in long periods. Please see detail in the papers below:

Hao Dong, Gary D Egbert, Divergence-free solutions to electromagnetic forward and adjoint problems： a regularization approach, 2019

Suzuki, Development of a three-dimensional magnetotelluric inversion program considering topography with cell-centered finite volume method, 2025

outputInterval : The output interval of iterations. If not set, It is output every iteration.

SettingEachLambda: The program adopts the damping parameter for model constraint get larger to smaller (cooling). Here, you set the parameters as below：

data [damping parameter] [step length] [ending criteria for the gradient error] [the number of maximum iterations] [ending criteria for the changing value from initial objective function] [ending criteria for the changing model parameters from previous ones]

decreaseFactor: In the inital iteration of each damping factor, when the objective function increases, the program reduces the step sise to multiply this value.

impedanceFile: The file name for Impedance Tensor. You should set the one in 1.Making Input Data from EDI file.

tipperFile: The file name for Tipper Tensor. You should set the one in 1.Making Input Data from EDI file.

Note that if you need restart the inversion, you can do it by coment-out the parameters below:

initialResistivityFile：You set the resitivity file for restart. It should be output like "Rho_[damping factor]_[the number of iterations].txt".

initialDistortionFile：You set the estimamted distortion file for restart. It should be output like "DistortionForRestart_[damping factor]_[the number of iterations].txt".

4.Execution of the Inversion.

You put the input files, obsDataImpedance.txt, obsDataTipper.txt, and the program (MT_INV_2024.exe) in the same folder, and then you can execute as below command:
MT_INV_2024.exe calcData.txt >info.txt"
">info.txt" can store the logs in the file "info.txt".
Note that you should have enough space in the storage in your computer, because many output file are created (So sorry...).
When you use the Australia data, we consider 64GB memory. If you are in less memory environment, please set parameter again for light calculation.
Furthe, this calculation will take time about several days with Intel® Core™ i9-10900. In our experiment, it takes about a day when you use 20k cells, ofcourse it depends on the other parameters...

5. Visualization

i. Visualization of Resistivity Model.

You can see the optimization log in "optimizeInfo.txt".
The resistivity models are output the files like below:
Rho_[damping factor]_[thenumber of iteration].vtk [or .txt]
You can visualize .vtk file with "Paraview".
In .txt files, the coordinates X,Y,Z, and resistivities in the center of the cells are output.
The figures below are the results using the Australia sample.

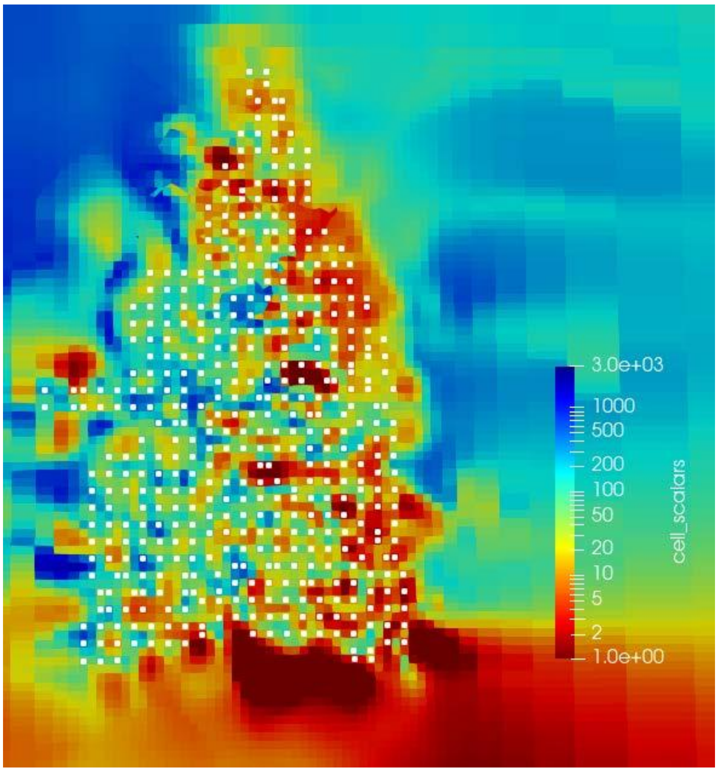

Figure. 6 Z = 200 m

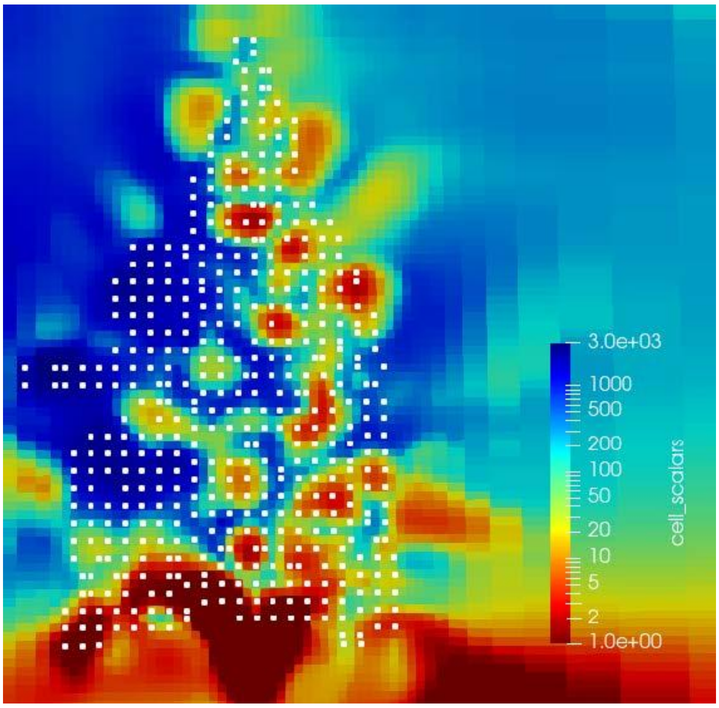

Figure. 7 Z = 1900 m

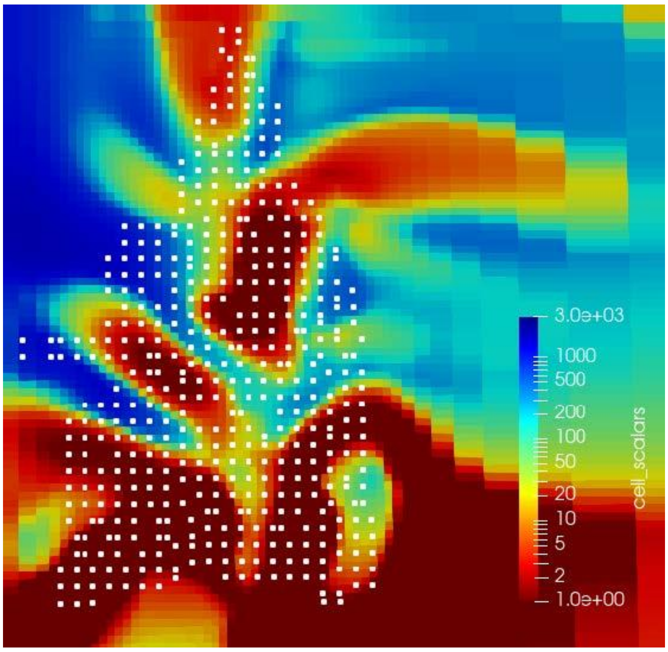

Figure. 8   Z = 7000 m

ⅱ. Plot the Obsservation and Calculation Curves.

Here, we plot response curves using "PlotCalcObsData.py".
First, you set parameters below.
The parameters are located in the line:

#===Parameters===================

The meanings of the parameters are as below:

obsCalcImpedanceFile: The calcculated impedance file. It is named like "ObsCalcImpedance_[damping factor]_[the number of itertation].txt

obsCalcTipperFile: The file of the obsercation of impedance tensor. It should be output when you run "readEdi.py".

ObsCalcTipperFile, obsTipperFile: These are the same as impedance files.

outputLocationFile: It is also output when you run "readEdi.py".

nFreqs: The number of frequencies. Please set the same value as you used in the inversion.

tipperPlot: You set "True" when you need to plot Tippers.

outputFolder: The folder for output files.

You can run the script as below:
python PlotCalcObsData.py

The sample picture is shown in below:

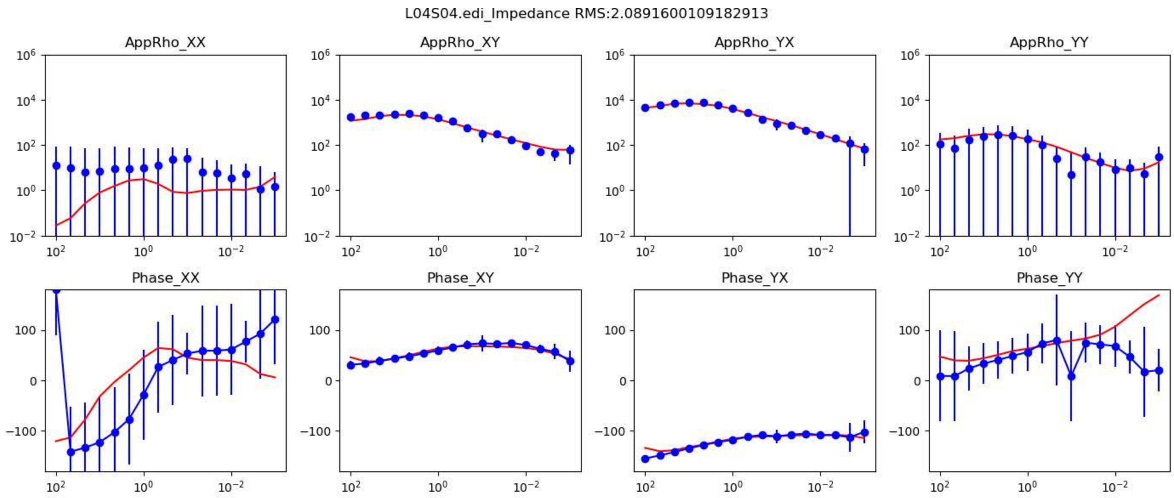

Figure. 9

6. Data format

We are now writing this section...

Please contact us with the email:
suzuki@tankai.co.jp or mk.pn14951011@gmail.com